

| Question number | Answer                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                                                                                                           | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(a)(i)         | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• because it is making predictions (1)</li> <li>• based on {knowledge / data} (1)</li> <li>• it is a visual representation (of what might happen) (1)</li> </ul> | <p><b>ACCEPT</b> guess / an idea / estimation / what might happen</p> <p><b>ACCEPT</b> statistics / what has happened</p> <p><b>NB</b> award {extrapolation / correlation / trend} once in either context of mp 1 or mp 2</p> | (2)  |

| Question number | Answer                                                                  | Additional guidance | Mark |
|-----------------|-------------------------------------------------------------------------|---------------------|------|
| 5(a)(ii)        | <ul style="list-style-type: none"> <li>33 / 33.3 / 33.33 (%)</li> </ul> |                     | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
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| *5(b)           | <p><b>1 Aspect : Positive impact on local (human) population:</b></p> <ul style="list-style-type: none"> <li>• crops will provide food</li> <li>• forests will provide materials for {building / making fires}</li> <li>• {crops / forest} will provide {employment / income}</li> <li>• {crops / forests} encourage new animals that might provide food</li> <li>• credit effect of reduced global warming on people e.g. no extremes of temperature, no droughts</li> </ul> <p><b>2 Aspect : Negative impact on local (human) population:</b></p> <ul style="list-style-type: none"> <li>• {crops / forests} may replace the crops they were growing already</li> <li>• {crops / forests} may {reduce / replace} the land they were using for animals</li> <li>• {crops / forests} may {reduce / replace} the land they were {living on / using for houses / factories}</li> <li>• crops use {pesticides / fertilisers} which is expensive for farmers</li> </ul> <p><b>3 Aspect : Positive impact on global warming:</b></p> <ul style="list-style-type: none"> <li>• carbon dioxide will be removed from the atmosphere</li> <li>• decreasing the greenhouse effect</li> <li>• extremes of temperature should be reduced</li> <li>• extremes of precipitation should be reduced</li> </ul> <p><b>4 Aspect : Negative impact on global warming:</b></p> <ul style="list-style-type: none"> <li>• planting crops may reduce carbon dioxide levels less than {forests / pre-existing crops}</li> <li>• machinery used in {planting {forests / crops} / harvesting crops} produces carbon dioxide</li> <li>• burning {plants / trees} produces carbon dioxide</li> </ul> <p><b>5 Aspect : Positive impacts on biodiversity:</b></p> <ul style="list-style-type: none"> <li>• planting {crops / forests} may increase (new) habitat</li> <li>• plants may provide food</li> <li>• increase in herbivores may increase carnivores</li> </ul> <p><b>6 Aspect : Negative impacts on biodiversity:</b></p> <ul style="list-style-type: none"> <li>• planting {crops / forest} could reduce (original) habitat</li> <li>• crops are monocultures and therefore reduce biodiversity</li> <li>• trees planted may not be suitable species</li> <li>• crops use pesticides that could harm food chains</li> </ul> | (6)  |

|         |     |                                                                                                                                                                                                                                                                                                                                                                                                           | Additional guidance                                                             |
|---------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Level 0 | 0   | No awardable content                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 |
| Level 1 | 1-2 | Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context. | Simple links made between plants and their impact                               |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.                                        | Extended links made between plants and their advantages or disadvantages        |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.         | Clear discussions on the types of plants and their advantages and disadvantages |

